

Supply chain sustainability in emerging economy: A negative relationship conditions' perspective

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ABSTRACT

Due to increasing reliance on suppliers, succeeding in supply chain sustainability hinges on the firm's collaborative relationship with its supply chain partners, positioning relationship conditions as a linchpin in firms' quest towards greater sustainability. This research scrutinizes if linkages between innovation, joint planning and resource sharing (i.e., JPRS or collaboration) and three dimensions of sustainability (i.e., supply chain, environmental, and social performance) are contingent on relationship conditions. Building on the force field theory and transaction cost economics, the moderation role of negative relationship conditions is examined using data collected from industries by deploying structural equation modeling (SEM) and SEM-based multi-group analysis. The relationship conditions include partner selection, power asymmetry, opportunistic behavior, and conflicting interests. Results show that the difficulty associated with partner selection played a considerable role in moderating the relationship between perceived interdependency and both collaboration and innovation. The influence of opportunistic behavior and conflicting interests to serve as a moderator in the relationships from innovation and collaboration to the three sustainability dimensions was tenuous, whereas the power asymmetry failed to serve as an influential moderator. This paper contributes to the planning and controlling supply chain sustainability literature by outlining paths to sustainability, considering negative relationship conditions within an emerging economy context, and offering pathways for economies of similar development to undertake these efforts.

1. Introduction

A sole focus of firms on economic performance in the past has been raising concerns on its sustainability, motivating them to heighten the importance attributed to social and environmental considerations. Recognizing social, environmental, and economic performance dimensions in unison is commonly referred to as the triple bottom line (TBL) (Elkington, 1994). Sustainable development, or sustainability based on the TBL, has been receiving considerable attention world-wide. To achieve true supply chain sustainability (SCS), implying good performance across the TBL and supply chain, supply chain members need to collaborate by executing joint planning and resource sharing (JPRS) due to the interconnectedness and dependencies of supply chain members (Chen and Kitsis, 2017; Hall and Wagner, 2012). These complexities contribute to the uncertainty that exacerbates bounded rationality within the context of sustainable supply chain management (Hall and

Wagner, 2012). Collaboration, coordination, and information sharing within this context can alleviate some of these uncertainties, leading to collaboration being acknowledged as a powerful strategy for achieving sustainable supply chain performance (Chen and Kitsis, 2017). Critical to keep in mind, though, is that the collaboration is not only founded on the JPRS but on building long-term relationships and culture based on trust, commitment, and loyalty (Kumar et al., 2018). With this as a foundation, collaboration can reduce bounded rationality and opportunism. In the absence of collaborative culture, JPRS and collaboration are used interchangeably in this study.

Another benefit of collaboration on sustainability initiatives can be greater innovation. Such innovation is needed to satisfy the increasing expectations of stakeholders also on both environmental and social dimensions (Dangelico and Pujari, 2010; Hall and Wagner, 2012; Pagell and Shevchenko, 2014). Innovation being enhanced by collaboration is in line with the imperative that innovation should not only be viewed

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within the firm's purview, but should also leverage supply chain partners' capabilities (Li and Nguyen, 2017). Within a network of supply chain partners, focal firms may develop innovation collaboratively (Seuring and Muller, 2008) by redirecting their shared resources, knowledge and skills—leading to new innovative products and processes that are cost-effective as well as environmentally and socially friendly. Capabilities of this nature have the potential to offer sustained competitive advantage. As such, we consider both collaboration and innovation as critical antecedents to TBL performance.

However, within this framing, we also recognize that the foundation for both collaboration and innovation may be more fruitful in some settings than in others. Specifically, we suggest that a greater perceived interdependency between the partners serves as a more favorable environment to foster collaboration and innovation. As such, when supply chain members realize that they are interdependent on each other for goals, resources, skills, and learning to achieve sustainable performance, they may be more motivated to engage in collaboration and joint innovation efforts, which we hypothesize to lead to greater SCS in turn.

While the attainment of economic and environmental performance through the pursuit of collaboration and integration has received considerable attention (Chen and Kitsis, 2017, 2017; Lee and Klassen, 2008; Sancha et al., 2016), only some studies (Arora et al., 2020; Luzzini et al., 2015; Wiengarten and Longoni, 2015) investigated the linkage between collaboration and all three sustainability dimensions empirically. However, most studies in this area investigate both collaboration and innovation aspects, suggesting them to serve as a panacea for sustainability; something which we emulate in the present paper. As such, literature finds green innovation to be positively related to sustainable processes and supply management (Gualandris and Kalchschmidt, 2014), economic performance, and competitive advantage (Gürlek and Tuna, 2018). Nevertheless, scholars also found evidence for the contextual nature of these relationships when some of their hypothesized relationships linking innovation to economic and environmental performance were not supported (Aguilera-Caracuel and Ortiz-de-Mandojana, 2013; Hall and Wagner, 2012).

Though supply chain collaboration is required for SCS, different kinds of collaborative relationship atmospheres (or conditions) exist in practice. A relationship atmosphere is an emotional setting representing two opposite poles of five elements—power/dependence, trust/opportunism, closeness/distance, cooperation/conflict, and expectations (Roehrich et al., 2002; Su et al., 2013). These elements have positive as well as corresponding negative aspects. A negative relationship atmosphere implies partners are opportunists, power is misused, conflicts among partners sustain, etc. In such negative relationship conditions, collaboration and innovation may become weak to yield desired results of improved SCS. Literature (Cheng, 2011; Huo et al., 2022; Michalski et al., 2017; Walter et al., 2015) highlights relationship atmosphere indicating relationship quality is critical in achieving success, i.e., sustainable performance, from collaborative relationship-based innovation. Extant literature (Cho et al., 2019; Li and Nguyen, 2017; Nyaga et al., 2013) highlighted the importance of relationship conditions and focused on positive aspects of relationship atmosphere elements. However, they do not sufficiently explain the linkages of collaboration-innovation-SCS and hardly pay attention to the linkages under different negative relationship conditions. If the elements of relationship atmosphere or conditions become more negative due to the evolving nature of businesses in emerging economies, contrary to the businesses of matured economies, the competitive and sustainability goals of the supply chain would be sabotaged. In emerging economies, where law enforcement is weaker, firms generally give more priority to economic outcomes than considering the environmental and social consequences of their operations. Further, emerging economies require collaboration and innovation (Kothiyal, 2018; Yourstory, 2020) to mitigate the problems attributed to poor infrastructure, poor quality, poor productivity, lack of latest technologies adaptation, uncompetitive supply chain, more

unorganized brick-and-mortar retail stores, more non-compliant to sustainability requirements, etc. (Maitra et al., 2020). The most pressing issues this study considers are certain negative relationship atmospheres (or conditions) that may hamper the initiation of collaborative and innovative relationships and, then, the influence of these collaboration and innovation on the SCS.

To ascertain certain negative relationship conditions as barriers, as described above, that could be conceptually related to our model; we reviewed related literature (Cheng, 2011; Cheng and Sheu, 2012; Cho et al., 2019; Huo et al., 2022; Kanyoma et al., 2020; Li and Nguyen, 2017; Michalski et al., 2018; Nyaga et al., 2013; Vollmer, 2015). It helped identify negative relationship conditions, such as power asymmetry, strategic partner selection, opportunistic behaviors, and conflicting interests, as relevant contingency variables to successful collaboration. These negative relationship variables were generally positioned as devaluing the relationship and increasing relational risks (Cheng, 2011; Michalski et al., 2017), thus serving as impediments to initiating and maintaining collaborative relationships for SCS. Therefore, building on the relationship (Roehrich et al., 2002; Su et al., 2013; Sutton-Brady, 2001) and supply chain literature (Huo et al., 2022; Kanyoma et al., 2020; Michalski et al., 2018; Touboulic and Walker, 2015), we consider power asymmetry, strategic partner selection, opportunistic behaviors, and conflicting interests as specific negative relationship conditions. Existing literature on the comprehensive impacts of these negative relationship atmospheres is limited and yet fragmented. To realize successful collaborative activities and its outcomes, it is imperative to understand how these elements of negative relationship atmosphere (i.e., relationship conditions) influence the relationships between collaboration, innovation, and SCS. To the best of our knowledge, these dynamics have never been investigated. Drawing from our logic and previous literature, we believe them to be critical to a better understanding of firms' quests toward greater sustainability initiation and maintenance. Specifically, this paper aims to answer the following research question.

RQ1. *How do relationship conditions, such as strategic partner selection, power asymmetry, opportunistic behavior, and conflicting interest, influence supply chain sustainability resulting from collaboration and innovation?*

In order to address the above research questions, a conceptual model is developed that draws on force field theory (FFT) and transaction cost economics (TCE). As our model has multiple abstract constructs, such as interdependency, collaboration, innovation, etc., we have developed and identified measurement variables for each construct. The model involving constructs representing multiple variables and their relationship (direct, mediation, and moderation) is the most appropriate for structural equation modeling (SEM), as it gives more confirmatory power. The model is subsequently tested with data collected from experienced respondents from diverse industries of different firm sizes; thus, our results are more generalizable. As moderation variables have multiple groups, SEM-based multi-group analysis was considered most appropriate to observe moderation effects.

By integrating FFT and TCE, this study offers a nuanced perspective by highlighting that collaboration-innovation-sustainability linkages are contingent on relationship conditions. We thus contribute to the literature that calls for more research into collaboration, innovation, SCS, and associated nuances (Hall and Wagner, 2012; Li and Nguyen, 2017; Neutzling et al., 2018; Zhu et al., 2012). To the best of our knowledge, this is one of the first studies to employ the FFT and TCE perspectives to develop a granular picture of developing and maintaining a sustainable supply chain management by analyzing if SCS is contingent on negative relationship conditions. This study also contributes to the literature that underscores the role of emerging economies in global business, their contributions to the environment and society, and the compelling need to understand sustainability and sketch a path to deploy effective sustainability strategies in those economies.

The rest of the paper is organized as follows. Section 2 outlines the

literature review. Section 3 develops the research model and underlying hypotheses. The methodology adopted in this study is presented in Section 4. Results and implications are discussed in Section 5. Finally, Section 6 concludes this study and also presents the future scope.

2. Literature review

This section discusses the relevant literature and highlights the gaps that this study aims to fill. Operations or SCP concerns with low cost, better quality, fast delivery, better flexibility (Wiengarten and Longoni, 2015), customer satisfaction, etc. Environmental performance is concerned with the minimization of harmful effects that operations can produce, the responsible use of natural resources, and the recycling of products (Carter and Rogers, 2008; Wiengarten and Longoni, 2015; Zhu and Sarkis, 2007). The social dimension is concerned with how supply chains influence society, including people within and outside of the firm. Specifically, it includes product safety, labor conditions, safe work environments (Carter and Rogers, 2008; Wiengarten and Longoni, 2015), and social welfare. A balanced performance across these three performance dimensions is popularly known as sustainability or TBL performance (Carter and Rogers, 2008; Elkington, 1994; Saputri and Lee, 2021).

2.1. Relationship conditions in collaborative sustainability

To have a productive supply chain relationship, literature (Chen et al., 2017; Fynes et al., 2005; Huo et al., 2022; Mohr and Spekman, 1994; Wang et al., 2022) has paid attention to the quality of buyer-supplier relationships. They considered trust, commitment, satisfaction, adaptation, and minimal opportunism as the main elements of relationship quality. The literature further found that relationship quality influences performance, product quality, service quality, and cost (Chen et al., 2017; Oyedijo et al., 2021; Wang et al., 2022). If risks in supply chain collaboration are high, partners will not trust each other, and consequently, relationship quality will not be good. Contrary to most previous studies, supply chain partners need to strive to reduce relationship risks to enhance trust or relationship quality rather than putting efforts into building trust to reduce relationship risks, as trust is developed over a long period (Kumar et al., 2016; Laeequddin et al., 2009). Prior research discussed the impact of relationship quality in terms of relationship strength, dynamic capability, coordination, organizational, technological, and environmental factors on collaborative relationship performance (Kumar et al., 2016, 2018; Wang et al., 2022). Their subsequent impacts on SCS are though not known.

The literature further advocates understanding the partner's characteristics as one of the important ways to reduce risks and increase the chance of buyer-supplier relationship success (Johnsen and Lacoste, 2016; Laeequddin et al., 2009; Mohr and Spekman, 1994). These characteristics include credibility (i.e., popular company), power, perception to exploit or opportunism, dependability, etc. Partner characteristics shape the relationship atmosphere that again builds partner characteristics. Relationship conditions or atmosphere is an emotional setting in which partners interact (Hedaa, 2007; Sutton-Brady, 2001). Relationship atmosphere with pairs of opposite roles is classified into five dimensions—power/dependence, trust/opportunism, closeness/distance, cooperation/conflict, and expectations (Roehrich et al., 2002; Su et al., 2013; Sutton-Brady, 2001). In 1982, the IMP group introduced the basic stepping stone for these dimensions constituting an interaction atmosphere (Håkansson, 1982). These dimensions, which are relatively mutually exclusive and collectively exhaustive, were refined and accepted as a framework for understanding the relationship atmosphere (Hedaa, 2007; Su et al., 2013; Sutton-Brady, 2001). A part of these dimensions, namely, dependence, trust, closeness, cooperation, and expectation, favor the relationship and improve relationship quality that determines the long-term relationship among partners (Wang et al., 2022). Their corresponding negative counterparts are power,

opportunism, distance, and conflict, which bring uncertainties in a collaborative relationship (Su et al., 2013; Zou and Wang, 2022) and are the under-investigated area in the industrial relationship (Johnsen and Lacoste, 2016). These uncertainties represent negative emotions of the relationship atmosphere (that may further culminate in anger, fear, sadness, guilt, etc.) which are more profound, impactful, and unvalued in literature (Hedaa, 2007; Koporcic et al., 2020). To reduce considerable risks to collaborative relationship, relationship arrangements should strive to understand and reduce these uncertainties (Laeequddin et al., 2009). As partner selection, power asymmetry, opportunism, and conflicting interests are the basic uncertainties in all relationships (Su et al., 2013; Sutton-Brady, 2001), this study further looks at the literature on these relationship uncertainties.

To achieve sustainability through collaboration, production, operations, and supply chain literature (Govindan et al., 2014; Jayaram and Avittathur, 2015; Kouhizadeh et al., 2021; Michalski et al., 2018; Oyedijo et al., 2021; Touboulic and Walker, 2015) marginally and fragmentally acknowledged the importance of several relationship conditions. Some of the major (negative) relationship conditions that act as barriers include power asymmetry (Huo et al., 2022; Michalski et al., 2017; Touboulic et al., 2014), partner selection (Barratt, 2004; Li and Nguyen, 2017), opportunistic behaviors (Touboulic et al., 2014; Zou and Wang, 2022), and conflicts (Barratt, 2004; Cheng and Sheu, 2012; Huang et al., 2020; Johnsen and Lacoste, 2016). The management of these barriers is thus critical for the successful pursuit of collaborative sustainability. We delineate these barriers below.

Partner selection is a critical element for collaboration, especially when innovation is one of the objectives (Corsten and Kumar, 2005; Dyer and Singh, 1998; He et al., 2016; Li and Nguyen, 2017; Srivastava et al., 2017). For example, He et al. (2016) highlight the importance of partner selection and provide guidance on how to do so within the agricultural product supply chain context. Various criteria can be applied in partner selection, ranging from strategic to operational (Li and Nguyen, 2017). Collaboration is able to generate relationship value when partners are willing to make relation-specific investments (Dyer and Singh, 1998), offering the foundation for creating inimitable advantages (Srivastava et al., 2017), especially in contexts where synergy-sensitive resources are critical (Dyer and Singh, 1998; Srivastava et al., 2017). However, challenges in distributing these values exist when one party is more powerful (Corsten and Kumar, 2005; He et al., 2016; Nyaga et al., 2013). In addition, if a partner is not trustworthy, the benefits derived from collaborative-innovation are at stake (Corsten and Kumar, 2005). Overall, literature has only, to a limited extent, explored the complexity of partner selection and its potential consequences, necessitating further study.

Power asymmetry can be characterized by gaps in negotiation power, financial position, and brand value. Firms with favorable power asymmetry can enforce sustainability codes, monitor weaker firms' operations, and assess desired sustainability performance better. As such, Touboulic et al. (2014) find that stronger firms can use power to drive sustainability and transfer risk and cost to weaker firms, which also attracts criticism. Literature (Boyd et al., 2007; Huo et al., 2022; Zou and Wang, 2022) describes how using power may bring frustration and tension that may destroy collaboration. Especially as small and medium-sized enterprises (SMEs) have limited bargaining power and resources (Lee and Klassen, 2008), exerting power over them may be detrimental. Further, Hall and Matos (2010) convey that power asymmetry subverts SSC as it hinders the acquisition and diffusion of knowledge. Power can be categorized as non-mediated or mediated. The former is relational and consists of expert and referent power, while the latter includes coercive, legitimate, and reward power (Benton and Maloni, 2005; Nyaga et al., 2013). Nyaga et al. (2013) studied the influence of non-mediated, mediated, and reward power on collaborative behavior and included power asymmetry as a control variable. Zhuang et al. (2010) find that the greater the firm's power, the more it will exercise noncoercive power, and the less likely it will exercise

noncoercive power. Power asymmetry or power imbalance is also often associated with less stable and less trusting relationships (Huo et al., 2022; Michalski et al., 2017). However, to the best of our knowledge, how power asymmetry moderates the linkage between perceived interdependency, collaboration, innovation, and SCS performance is unknown.

Opportunistic behavior can also be a significant impediment to sustainable collaboration-innovation relationships. It is defined as self-interest seeking with guile, leading to deceit-oriented behavior and a violation of implicit or explicit promises (Morgan and Hunt, 1994; Sancha et al., 2016). Researchers (Cheng and Sheu, 2012; Morgan et al., 2007; Morgan and Hunt, 1994; Nyaga et al., 2013; Sancha et al., 2016; Walter et al., 2015; Zou and Wang, 2022) have highlighted the pitfall of opportunistic behaviors in supply chain partnerships from multiple angles. If partners' capacity, information, or resources are imbalanced, powerful partners may behave opportunistically and appropriate greater value in the relationship (Huo et al., 2022; Nyaga et al., 2013; Zou and Wang, 2022). Due to information asymmetry, the supplier may also use unsustainable practices, such as opportunism (Huo et al., 2022; Sancha et al., 2016). This behavior can be reduced by supplier (or buyer) assessment and monitoring (Carter and Rogers, 2008; Sancha et al., 2016). Cheng and Sheu (2012) find that opportunistic behavior negatively moderates the link between relationship orientation and strategy quality. Morgan et al. (2007) note that the opportunistic behavior of focal suppliers reduces category management performance and promotes militant behaviors of non-focal suppliers. Walter et al. (2015) find that extensive formalization promotes opportunistic behaviors but that communication quality alleviates opportunism. At the same time, most literature highlighted the pitfalls of opportunistic behavior but did not provide empirical evidence, especially within a sustainability context, motivating us to include this variable in our investigation.

Conflicting interests are reflective of disagreement on issues of mutual importance and have the potential to reduce cooperation, satisfaction, and performance (Zhuang et al., 2010). Partners may not aim exactly for the same goals, leading to conflict in the relationship (Huang et al., 2020). How conflicts are managed impacts collaboration (Johnsen and Lacoste, 2016; Oyedijo et al., 2021). Literature (Benton and Maloni, 2005; Cheng, 2011; Cheng and Sheu, 2012; Eckerd and Sweeney, 2018; Johnsen and Lacoste, 2016; Nyaga et al., 2013; Oyedijo et al., 2021; Vollmer, 2015; Zhuang et al., 2010) has highlighted the negative consequences of conflicting interests in collaboration and suggested ways to address it. The types of conflicts are multifarious, including task, relationship, and process conflicts (Vollmer, 2015). Some studies have modeled task conflict as an independent variable and innovation as the dependent variable and found a positive relationship between the two. However, other studies find relationship conflict to be negatively related to innovation (Vollmer, 2015). In addition, Cheng and Sheu (2012) find dysfunctional conflict to be negatively moderating the link between relationship orientation and strategy quality. Conflict may result from opportunistic behavior and power asymmetry (Cheng, 2011). However, asymmetric power also potentially being the cause of conflict, dissension, and dissatisfaction (Benton and Maloni, 2005; Johnsen and Lacoste, 2016), while at the same time being able to be used to resolve conflict (Belaya et al., 2009; Nyaga et al., 2013). Literature (Cheng, 2011; Eckerd and Sweeney, 2018) considers conflict as a major relational risk and finds that it reduces the extent of knowledge sharing. To an extent, it may also be a driving force and creativity (Johnsen and Lacoste, 2016). Zhuang et al. (2010) find that exercising coercive power increases conflict, which in turn reduces cooperation. To alleviate this situation, partner selection processes and collaboration allow to align expectations, goals, and behaviours that will minimize conflict and misunderstanding (Nyaga et al., 2013). Overall, literature suggests conflict as being detrimental for relational rent and reducing performance, making as a critical contingency to be considered within our context.

Relationship conditions such as power asymmetry, partner selection,

opportunistic behavior, and conflicting interests can act as critical impediments to collaboration and innovation for sustainability.

2.2. Studies in emerging economies

Emerging economies are playing a considerable role in the production of goods and provision of services, which are not only then consumed in these markets, but also exported to the developed world. The emerging economies index suggests that emerging markets account for more than 10% of the world's market capitalization, heightening the need to achieve TBL performance (Jayaram and Avittathur, 2015; Katiyar et al., 2018). India is ranked as one of the top destinations of foreign direct investment (Katiyar et al., 2018) and has been noted as one of the fastest-growing economies (Mishra, 2018). This makes India an important economy in implementing sustainability.

Firms in emerging countries play a crucial role in global supply chains, pushing these countries like China, India, and Brazil to make inroads into SCS (da Silva et al., 2013; Jayaram and Avittathur, 2015; Katiyar et al., 2018; Zhu and Sarkis, 2007). For instance, managing the supply chain's environmental aspects is reported to positively affect environmental and economic performance in China (Zhu and Sarkis, 2007). In Brazil, collaboration with stakeholders, power, and mutual dependence is found to be important for sustainability (da Silva et al., 2013; Silva et al., 2021). A study (Jayaram and Avittathur, 2015) in India finds that sustainability strategy (i.e., collaboration) improves product recovery and recycling. Another study (Katiyar et al., 2018) based in India finds careful implementation of supply chain functions, planning, manufacturing, delivery, and sourcing to drive sustainable performance, which improves supply chain performance. However, very few studies exist that trace and suggest paths to sustainability and its dynamics in the presence of relationship conditions in emerging economies.

As such, the exploration of barriers to collaboration and sustainability may be even more important in these contexts. Therefore, it is critical to unearth how these relationship conditions moderate the dynamics of initiating, maintaining, and sustaining supply chain partnerships for sustainability.

It is evident from the literature described above that emerging economies require more attention when aiming to understand the perceived interdependencies that motivate firms to pursue collaboration and innovation in their quest toward sustainability performance. As such, this study attempts to understand interdependency-collaboration-innovation-TBL dynamics, including barriers that may hamper this quest.

2.3. Literature gaps

It is evident from the above discussion that extant literature widely acknowledges the negative aspects of the relationship conditions or atmosphere, such as power asymmetry, opportunism, distance, and conflict present in the relationship. A broad tendency in the literature is to focus on positive relationship atmosphere elements, such as trust, commitment, cooperation, common goals, etc., that favor and strengthen the relationship atmosphere (Johnsen and Lacoste, 2016; Koporcic et al., 2020; Oyedijo et al., 2021). However, literature on the interplay of negative aspects of uncertainties creating relationship stress and their comprehensive impacts is limited (Koporcic et al., 2020; Zou and Wang, 2022). They are further fragmented in different contexts. A focus on non-rational negative behaviors of partner firms (and human behaviors) could enrich and enhance relationship-based research (Koporcic et al., 2020; Oyedijo et al., 2021) on operations and supply chains. Despite the increasing attention on collaboration for sustainability as a key factor to achieving competitive advantage, it is hardly known in the literature how the above uncertainties play out for sustainability performance in collaborative relationships. These negative aspects of the relationship atmosphere need to be further developed and

understood for sustainable collaboration. To address these literature gaps, it is required to investigate and understand the interplay of these negative uncertainties in the relationship for sustainability that can help better manage the relationship beyond its dissolution. As partner selection, power asymmetry, opportunism, and conflicting interests are the basic negative uncertainties in all relationships (Johnsen and Lacoste, 2016; Su et al., 2013; Sutton-Brady, 2001), this study attempts to go deeper into these uncertainties and uncovers their intricacies in sustainable collaboration.

3. Research model

The overall research model is presented in Fig. 1. The model suggests that a firm's perceived interdependency motivates collaboration and innovation, which in turn yields SCS, i.e., performance across the three TBL dimensions. We further suggest that within this process, firms face two sets of relationship conditions-based barriers or atmosphere: one before the collaborative relationship (i.e., Interdependency- > Collaboration and Interdependency- > Innovation) and another one during the practice of the collaborative relationship (i.e., Collaboration- > TBL Performance and Innovation- > TBL Performance). As circumstances surrounding firms and collaborative relationships are different in the two phases, firms face barriers of different natures in the two phases. However, some barriers or impediments may be relevant in both phases. Power asymmetry is relevant in both phases, i.e., before and after initiating collaboration. Firms would think about whether to collaborate with stronger firms because, during the collaboration, stronger firms may influence sustainable collaborative activities more. Firms face the situation of partner selection before starting a collaborative relationship. However, partners can experience opportunistic behaviors and conflicting interests of partner firms more realistically during the practice of collaborative relationships. Thus, we have different relationship conditions or barriers in the two phases of Fig. 1.

The FFT and TCE give us a ground to include barriers (i.e., negative

relationship atmosphere) in the model in two stages. Both theories highlight the existence of various relational and organizational resistances that prevent supply chain members from improving their competencies. The former theory is concerned about the resistances that prevent changes in the existing processes and working styles (Dent and Goldberg, 1999), while the latter relates to the barriers that make existing processes less effective in achieving desired results. Our model integrates both theories in explaining the role of relational conditions in achieving SCS from the interdependence of supply chain members for SCS and collaborative relationships. Both theories are briefly discussed below.

3.1. Force field theory

The change management needed by firms entering into sustainable collaboration can be viewed from the lens of Lewin's (1951) classic FFT (Soosay and Hyland, 2015). When firms realize that they are interdependent and have aligned goals (i.e., sustainability), they undergo collaboration (Soosay and Hyland, 2015) and innovation for sustainability (see Fig. 1). At this stage, FFT is relevant to our model that says some restraining forces oppose firms undergoing collaboration and innovation, i.e., change in firms' state (Kouhizadeh et al., 2021; Soosay and Hyland, 2015). According to the theory, firms exist in an equilibrium state until an external force, such as stakeholders' pressures, regulatory requirements, social responsibilities, or changing market dynamics, makes the change inevitable. When firms undergo change (sustainable collaboration in our case), several forces can restrain the change (Dent and Goldberg, 1999; Kouhizadeh et al., 2021). Here, two forces are at play: the driving and restraining forces (Soosay and Hyland, 2015). The driving force drives firms towards the required change, while the restraining force opposes the change. The balance between driving and restraining forces determines the outcome of the transition phase. Firms attempt to prevail by putting pressure on those who oppose them. When the driving force is stronger than the restraining force, firms

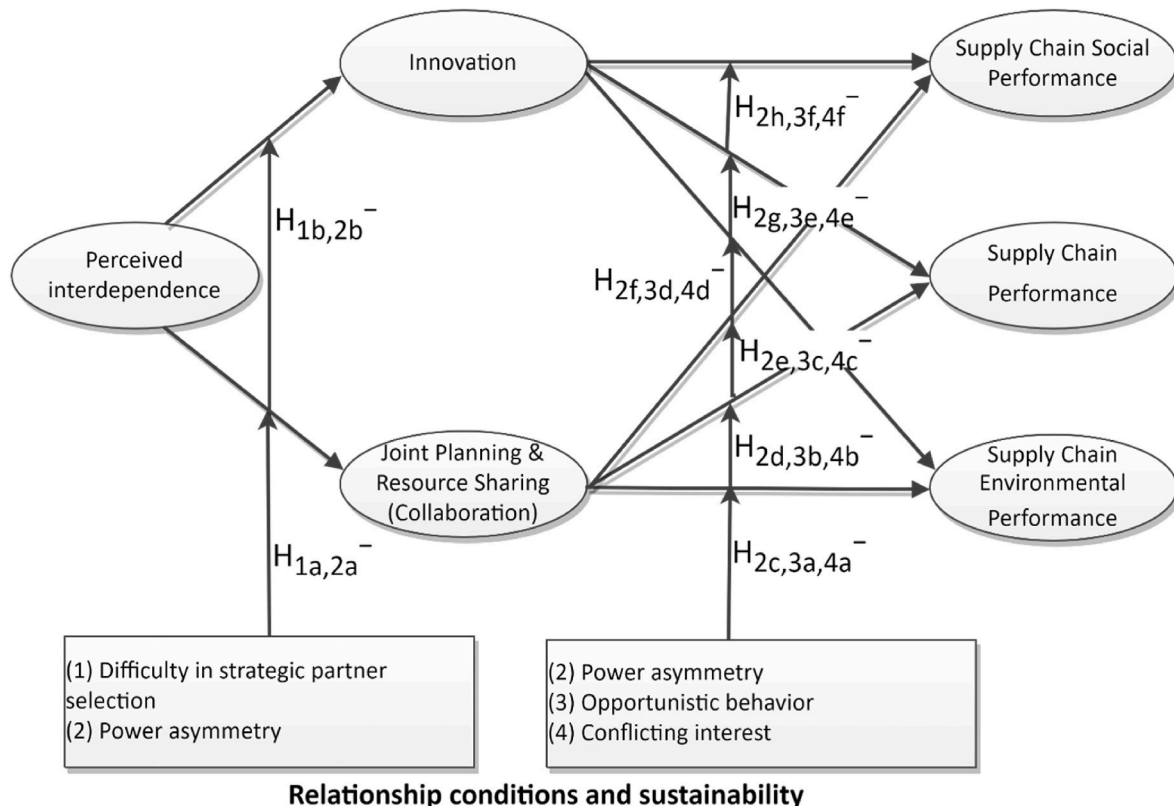


Fig. 1. Conceptual model (our main focus is to test moderation effects and direct relationships are considered as a baseline; hence they are not hypothesized here).

realize the need for change and enter another equilibrium state. In our model, barriers to sustainable collaboration act as a restraining force that oppose the change (i.e., entering into a collaborative relationship). When firms understand their interdependency on other supply chain firms, they put efforts to enter into collaboration and innovation, and here some barriers act as restraining forces and obstruct firms from entering into collaboration and innovation for SCS. These barriers include power asymmetry and (difficulties in suitable) partner selection in this study (see Fig. 1).

We suggest that power asymmetry plays a role in both stages, i.e., firms entering into collaborative relationships and then leveraging that collaboration for SCS. In addition, partner selection is suggested only for the transition into collaboration since that choice is mute once the collaboration has started.

3.2. Transaction cost economics

Collaboration for SCS can be a transaction-specific investment that comes at a cost. Transaction costs include the cost of managing the relationship and opportunity costs. Opportunity costs may be too high under certain conditions of poor governance, inefficient mutual interactions, and self-interest (Williamson, 2008). Self-interest may include lying, opportunistic expropriation, fostering conflicts, leveraging power asymmetry, cheating, and violating the agreements.

Asset specificity, uncertainty, and governance mechanisms are primary constructs that have a fundamental representation in TCE (Grover and Malhotra, 2003). Asset specificity represents the degree to which assets have value specific to the transaction or relationship. A greater degree of asset specificity indicates that assets committed to the relationship have little value outside the relationship and therefore represent costs and introduce risks, such as possible hold-ups or opportunism (Grover and Malhotra, 2003). Uncertainty refers to unanticipated sources of disturbances surrounding the transaction. The uncertainty limits contract formulation ex-ante and/or the ability to verify compliance ex-post (Grover and Malhotra, 2003). Greater uncertainty invites additional coordination costs and risks of performance failure. Governance mechanisms refer to the flow of materials and services and how they are directed through the process steps (Grover and Malhotra, 2003). TCE aligns individual transactions with modes of governance to minimize transaction costs (Patrucco et al., 2020; Williamson, 2008). Williamson recognizes three forms of governance: market (i.e., discrete transactions), hierarchy (i.e., authoritarian-based command and control arrangements), and hybrid (i.e., between market and hierarchy). Blomqvist et al. (2002) describe hybrid governance as a continuum of partnership options that reflect a buyer-supplier collaborative relationship driven by trust, commitment, and endurance (Patrucco et al., 2020) that are instrumental in reducing transaction costs. Collaboration or strategic alliance, vertical value chains, cooperative behavior, contracts, franchising, etc., are hybrid of market and hierarchy forms of governance. Asset specificity, uncertainty, and frequency of transactions are intermediate in a hybrid form of governance that is more adaptive and capable of dealing with unforeseen disturbances (Koo et al., 2020; Williamson, 1985). Opportunism, asymmetry, and uncertainty may also exist in this mode of governance, but it will yield good expected value as the governance is exercised for a long duration.

To establish effective governance that can minimize transaction costs—collaboration—a relationship-based governance mechanism, which stands in contrast to contract-based governance, is suggested. We consider collaboration and collaborative innovation as relational arrangements. Collaborative innovation is innovation in collaboration with supply chain partners. It also implies collaboration and innovation. The collaboration is governed by the relationship atmosphere that has its positive as well as negative aspects (Su et al., 2013; Sutton-Brady, 2001). A negative relationship atmosphere brings uncertainties in the collaboration that prevent realizing the expected outcomes of sustainability performance. It develops a different kind of governance structure

that may not support how collaboration should work, and sustainable performance outcomes are expected. A negative relationship atmosphere works against the trust, commitment, and cooperation in the relationship (Su et al., 2013), and thus it brings high transaction costs that may sabotage realizing better sustainability performance. Against the backdrop of such relational governance, we aim to study how a negative relationship atmosphere (or various negative collaborative relationship conditions) works against the relationship between both collaboration and innovation and SCS performance. Collaboration, innovation, and negative relationship atmosphere (i.e., relationship conditions) are anchored in the three core constructs of TCE, namely governance, assets specificity, and uncertainty, respectively. Specifically, we consider three elements of a negative relationship atmosphere: power asymmetry, opportunistic behaviors, and conflicting interests.

Research hypotheses are developed in Fig. 1 and motivated by FFT and TCE. We acknowledge that direct links between collaboration, innovation, and TBL performance are not entirely new. Hence, our focus is to test the moderation hypotheses.

3.3. Moderation effects

Resisting forces may be at the individual firm level, supply chain level, or external environment level, and these three types of resisting forces can impact supply chain collaboration (Fawcett et al., 2008; Oyedijo et al., 2021). We consider partner selection as prominent resistors at the individual firm level; power asymmetry, opportunistic behavior, and conflicting interest as another set of prominent resistors at the supply chain relationship level.

3.3.1. Difficulty in strategic partner selection

Perceived interdependency is what a firm thinks about its interdependency on the other (supply chain) firms (Kumar et al., 1995). When a firm thinks they depend more on the other supply chain partners for SCS, perceived interdependency would be high. New business requirements, such as the need to be more sustainable and collaborative, can trigger firms to transform their business strategy. The process of transforming business strategy to build new capabilities engenders resistance (Fawcett et al., 2015). As such, selecting the right partners for collaboration is a prerequisite and one of the most challenging tasks when aiming to initiate collaborative relationships. Drawing from FFT, having a strategic partner is a major hurdle that can prevent firms from undergoing the needed transformation from a state without collaboration to a state with collaboration. Partners with common goals (such as the dedicated pursuit of sustainability) and complementary capabilities (such as R&D and innovation) (Corsten and Kumar, 2005) are justifications for collaboration since it enables them to create value (i.e., relational rents) that cannot be created independently. To generate relational rents, partners should be able to (1) invest in relationship-specific assets, (2) use effective governance mechanisms, (3) exploit knowledge-sharing approaches, and (4) “use complementary capabilities (Dyer and Singh, 1998). Such partners are strategic partners. In addition, a minimum level of trust, commitment, communication (Mohr and Spekman, 1994), and loyalty are required to exploit these abilities and reduce the transactional costs of JPRS and innovation (Kumar et al., 2016). However, identifying partners and assessing them for these abilities and attributes are significant challenges, preventing JPRS and joint innovation activities. As such, potential strategic partners need to be assessed across strategic, tactical, and operational levels (Li and Nguyen, 2017). Depending upon prior experience, internal search, and an evaluation of capabilities, firms vary in their ability to identify and assess potential partners (Corsten and Kumar, 2005; Dyer and Singh, 1998).

Potential strategic partners must also be understood in terms of their potential for long-term trust-building opportunities, the existence of conflicting interests, potential opportunistic behaviors, their respective learning attitudes, and their sincerity in settling conflicts. In addition,

the deployment of conflict resolution mechanisms (Mohr and Spekman, 1994) and performance measurement techniques are essential. Clearly, challenges associated with selecting the right partners for long-term collaborative relationships will impact firms in JPRS and innovation initiatives in a negative fashion. Hence, we hypothesize the following.

H1a— Difficulty in strategic partner selection negatively moderates the relationship between perceived interdependency and JPRS.

H1b— Difficulty in strategic partner selection negatively moderates the relationship between perceived interdependency and innovation

3.3.2. Power asymmetry

There are natural barriers to inter-organizational relationships, with power asymmetry being one of them. Power asymmetry can deter firms in both stages: first, when firms prepare to enter into a collaborative relationship, and second, when firms are in such a relationship. We thus theorize about the role of power asymmetry in both stages.

First, firms can face resistance caused by power asymmetry when entering collaborative relationships. In inter-organizational relationships, a more powerful firm can dictate terms in its favor, undermining relationship-based long-term collaboration. Drawing from FFT and TCE, power asymmetry is counterproductive for firms aiming to enter into a collaborative relationship. According to FFT, when firms witness changes in their state, power asymmetry can effectuate the resistance to changes, especially the move toward more JPRS. With greater power imbalance, the weaker partner may be more insecure about the benefits-sharing mechanisms in collaborative settings. As such, weaker firms may fear that more transaction costs and risk could be transferred to them (Cho et al., 2019), and they may not benefit from the collaborative relationship (Touboulic et al., 2014).

Similarly, the gains from collaboration and innovation may be shared disproportionately. The weaker firm may be cautious when excessive pressure is imposed upon them, and the collaborative relationship may suffer. If the supplier is larger and more powerful, the buyer may suspect that the supplier may act opportunistically, which will prevent active collaboration and innovation to improve TBL's performance. Stronger players may also speculate whether their partners are able to deliver enough value, contribute enough resources, and share knowledge freely. As such, power asymmetry is expected to deter the honest resource commitment required for collaborative relationships. In such situations, firms' willingness to adapt or collaborate will be reduced. Thus, we hypothesize.

H2a— Power asymmetry negatively moderates the relationship between perceived interdependency and JPRS.

H2b— Power asymmetry negatively moderates the relationship between perceived interdependency and innovation.

Second, while being in collaborative relationships, power asymmetry can lead to compromised TBL performance by reducing the effectiveness of the collaborative relationship (Touboulic et al., 2014). Threatened by the imbalanced use of power, weaker firms may leave the relationship and look for new partners. As such, when the weaker partner invests resources and capabilities in the relationship, they may be concerned about the possibility of the stronger partner threatening to terminate the relationship if certain demands are not fulfilled, essentially using their investment made as a hostage. Such concerns may be prevalent even when the stronger firm also invests in the relationship. In this case, the powerful partner could exert coercive pressures to make the weaker partner invest more heavily in innovation, environmental, and social causes, rendering great potential for the relationship to get damaged.

Power asymmetry will likely yield opportunism, as the stronger partner may use its advantage to appropriate greater value (Nyaga et al., 2013; Touboulic et al., 2014). For instance, one study showed that while suppliers of major retailers experience improved performance in the relationship, suppliers have a greater degree of inequity (Corsten and

Kumar, 2005). As per TCE, such opportunism and asymmetric behavior of partners make collaborative and innovation activities ineffective and inefficient, preventing firms from achieving TBL performance.

The asymmetric relationship is also a beneficial attribute in collaborative relationships to effectuate supplier compliance (Michalski et al., 2017). This would be the case only when a stronger partner's intentions are not self-interest driven but novel and responsible. However, a strategically asymmetric relationship is not always favorable (Belaya et al., 2009; Li and Nguyen, 2017; Nyaga et al., 2013) since a powerful partner can undermine the balance of power and act opportunistically. As such, the power imbalance is often perceived as having a negative impact on collaborative relationships, rendering decreased TBL performance (Touboulic et al., 2014). For example, weaker firms may not have effective mechanisms to monitor or influence stronger ones and are more likely to comply with stronger ones for fear of losing business (Nyaga et al., 2013). More importantly, it can become a hurdle to long-term JPRS and innovation when powerful firms do not value their partners' views, do not lend the required support, or share rewards honestly. In such cases, power asymmetry will be counter-productive for JPRS and innovation to achieve sustainable TBL performance. Hence, we hypothesize that.

H2c— Power asymmetry negatively moderates the relationship between JPRS and SCEP.

H2d— Power asymmetry negatively moderates the relationship between JPRS and SCP.

H2e— Power asymmetry negatively moderates the relationship between JPRS and SCSP.

H2f— Power asymmetry negatively moderates the relationship between innovation and SCEP.

H2g— Power asymmetry negatively moderates the relationship between innovation and SCP.

H2h— Power asymmetry negatively moderates the relationship between innovation and SCSP.

3.3.3. Opportunistic behavior

Collaboration is an arena for potential opportunistic behavior by partners with different sets of goals. Opportunistic behavior may dilute the primary purpose of sustainable collaborative innovation, which is to create unique sustainable values. For instance, in order to reduce costs, suppliers may employ child labor, not maintain healthy work conditions, or mandate employees to handle products without appropriate protection (Sancha et al., 2016). Such behaviors require greater supplier assessment and audit, thus, increasing transaction costs and leading to compromised SCSP. From a TCE lens, such opportunistic behavior has the potential to diminish the effects of collaboration and collaborative innovation.

Specifically, under opportunistic behavior, the cost of JPRS and joint innovation may become too high, deterring firms from further resource commitments, and thus reducing the impact on TBL performance. To prevent opportunistic behavior, partner firms need to deploy mechanisms to safeguard their interests, which will increase transaction costs. In exchange relationships, opportunism limits relation-based governance (Nyaga et al., 2013), compromising the potential benefits of collaboration and innovation. For critical resources and capabilities, an imbalance in dependency is a desirable outcome for the more powerful partner (Morgan et al., 2007), who may act in their own short-term interest. Since this opportunistic behavior goes against the primary goal of collaboration, it compromises the speed and effectiveness of innovation, learning, knowledge creation, and resource optimization, leading to reduced SCSP, SCP, and SCEP. Perceived opportunistic behaviors can also lead to a decrease in trust and commitment to sustainability, curtailing JPRS.

If a partner's behavior is inconsistent with prior contracts or

agreements, they are said to behave opportunistically (Cheng and Sheu, 2012). For example, when joint innovation creates new skills and knowledge, one partner may leak these to obtain short-term benefits. Such behavior compromises the desired competency in achieving SCEP, SCP, and SCSP. As such, since partners need to protect their information, skills, and tangible resources against such behavior, this may undermine their competencies. If the level of opportunistic behavior is high, it will subvert JPRS goals, and firms may not engage in a further relationship, thereby compromising TBL performance. Thus, we hypothesize the following.

H3a— Opportunistic behavior negatively moderates the relationship between JPRS and SCEP.

H3b— Opportunistic behavior negatively moderates the relationship between JPRS and SCP.

H3c— Opportunistic behavior negatively moderates the relationship between JPRS and SCSP.

H3d— Opportunistic behavior negatively moderates the relationship between innovation and SCEP.

H3e— Opportunistic behavior negatively moderates the relationship between innovation and SCP.

H3f— Opportunistic behavior negatively moderates the relationship between innovation and SCSP.

3.3.4. *Conflicting interest*

Conflicting interests or conflict can harm resource use, leading to the distortion of information (Cheng and Sheu, 2012) and inhibiting integration (Kanyoma et al., 2020). For example, relationship conflict emerges when a team's goals are misaligned (Vollmer, 2015). Within our context, firms may not consider all three dimensions of TBL performance equally important and assign different importance levels to each. This may lead to conflicting interests in the allocation of scarce resources. While one partner may prefer to enhance environmental performance, the other partner may want to emphasize social performance. This conflict may be intensified if partners are from different cultures and/or when they are emotionally distant, subsequently reducing informal and open interactions. Open communication and a good collaborative culture (Kumar et al., 2016) are required to reduce conflict and enhance performance. Otherwise, such conflict will reduce the effectiveness of JPRS and innovation in realizing TBL performance.

In line with TCE, firms collaborate to lower transaction costs. However, conflicting interests increase the cost of implementing JPRS and innovation. The use of power in collaboration can yet yield different effects. While the use of coercive power to ensure sustainability compliance is a major reason for conflict in relationships, this power can also be used to coordinate and resolve conflict that can enhance SCP (Belaya et al., 2009; Nyaga et al., 2013). If power is not used coercively, relationships have the potential to improve.

Partners often have perceptual differences leading to conflict, dissatisfaction, and disproportional gain distributions (Nyaga et al., 2013). Nonaligned partners' behavior and expectation can result in conflict and misunderstanding, negatively impacting joint activities. To resolve conflicts, conflict resolution techniques need to be deployed (Mohr and Spekman, 1994), which can increase the cost of engaging in collaborative relationships and influence both relationship quality and performance (Touboulic and Walker, 2015). As such, TCE encourages firms to engage in collaborative relationships to eliminate transaction costs; however, the emergence of conflict sabotages the partners' interactions and increases transaction costs associated with JPRS and innovation. Hence, we hypothesize the following.

H4a— Conflicting interest negatively moderates the relationship between JPRS and SCEP.

H4b— Conflicting interest negatively moderates the relationship

between JPRS and SCP.

H4c— Conflicting interest negatively moderates the relationship between JPRS and SCSP.

H4d— Conflicting interest negatively moderates the relationship between innovation and SCEP.

H4e— Conflicting interest negatively moderates the relationship between innovation and SCP.

H4f— Conflicting interest negatively moderates the relationship between innovation and SCSP.

4. Methodology and results

We reviewed the literature related to the relationship aspects of SCS and identified the research question. We mainly used Scopus and Google Scholar search engines to select the relevant literature. In addition, we also used search engines of ScienceDirect, Emerald, Taylor & Francis, Wiley, and Informa separately. Different combinations of keywords were used to search the relevant articles. Keywords used were relationship quality, relationship characteristics, relationship conditions, collaboration, buyer-supplier relationship, supply chain, innovation, and sustainability. We filtered relevant articles by looking at their title and abstract from search results. Also, some of the most cited important and relevant studies are selected from these relevant articles. The research question was formulated after reviewing these relevant selected articles. A conceptual model was developed to answer the research question involving different linkages between constructs and the effects of relationship conditions. The constructs were measured by multiple items that were identified from the literature. Few items were added and modified to suit the context of this study. Based on these items, we designed a questionnaire by following standard guidelines for questionnaire design and related literature.

Interdependency as a reason to go for collaboration and innovation to achieve SCS was measured by dependency on skills, green initiatives, and effective solutions (Zacharia et al., 2011). Innovation was measured by utilizing learning together for new solutions (Azadegan et al., 2008), developing environmentally or socially friendly products/services on time, and providing freedom to create new procedures/tasks (Calantone et al., 2002; Gualandris and Kalchschmidt, 2014). JPRS was measured by eco-friendly product development (Youn et al., 2013), material requirement planning combined with recycled materials (Kumar et al., 2016), purchasing with green supplier assessment, reduce, reuse, and recycle (3 R) practice (Azevedo et al., 2011), end-users environmentally oriented demands, technology and machinery sharing (Handley and Benton, 2012), and inventory related information (Kumar et al., 2016). SCEP was measured by waste reduction, compliance with laws, increased recycling, and overall environmental performance (Large and Thomsen, 2011). SCP was measured by shorter lead time, improved quality, higher profit, and enhanced reputation with customer satisfaction (Kumar et al., 2016). SCSP was measured by community health and safety, better working conditions, ability and willingness to help (donation, resource) (Paulraj et al., 2014; Sancha et al., 2016), and always giving true information (Azzi et al., 2012; Lu et al., 2012; Yuen et al., 2017). Moderator variables were assessed using a single-item variable.

Experts having experience in similar domains reviewed the questionnaire for ambiguity, readability, and clarity. The questionnaire contained questions on the demographic profile of respondents, one question for each item, and a question asking their subjective opinions on the subject. Due to the infant stage of sustainability in emerging countries, difficulties in obtaining data, and the potential to misunderstand survey items, convenience sampling was used (Patrucco et al., 2019; Zhu and Sarkis, 2007), including those respondents who were approached through direct and indirect contact. Using Google form the survey was conducted to collect data from companies in India (Kumar

et al., 2018). We assured the confidentiality of the data, conveyed that their opinions and experience matter in this study, and promised a complimentary research report as an incentive to participate in this study. The survey was conducted in two phases. In the second phase, the questionnaire was sent to those who were not asked for a response in the first phase. After removing four responses with excess missing values, 119 valid responses in the first phase and 52 valid responses in the second phase, yielding 171 responses for further analysis. To determine whether the data collected in the two phases differed significantly, we conducted Chi-square difference tests, which turned out to be not significant ($p > 0.05$) for all variables.

Respondents had the titles of manager, consultant/analyst, director, and vice president. They were from the automobile (19.3%), software (18.7%), electrical/electronics/computer (11.1%), and other industries (50.9%). 57.9% of respondents were of age less than and equal to 30 years, 38% of them were between 30 and 50 years, and 3.5% were older than 50 years. We had 64% of firms with employees more than 1000, 23% between 500 and 1000, and the rest had less than 500 employees. Respondents' firms with 57.3% of the respondents collaborating with both suppliers and customers; 29.8% collaborated only with customers, while the remainder collaborated only with suppliers (service providers, manufacturers, wholesalers). It can be noted that software companies require to develop sustainable software products (Saputri and Lee, 2021), where collaboration and innovation would be critical for them.

Only select items were modified or added to suit the context of emerging economies. As almost all items have literature support, we assured content validity. Items were measured on a five-point Likert scale, ranging from 'poor' or 'strongly disagree' (1) to 'excellent' or 'strongly agree' (5). First, we tested the measurement model by employing exploratory and confirmatory factor analysis (CFA) to test hypotheses. As the model involves constructs measured by multiple items and a structure of paths connecting multiple constructs, the SEM approach was deployed to test the underlying path coefficient and its significance level.

4.1. Measurement model

An exploratory analysis was conducted using SmartPLS (Ringle et al., 2015) and SPSS. Results conveyed all (except one loading of 0.605) loadings greater than 0.70. One item of the perceived interdependency scale had a loading less than 0.60, which was thus removed from the scale. The average variance extracted (AVE) and the composite reliability (CR) of all constructs are greater than 0.50 and 0.70, respectively. Cronbach's alpha of constructs ranges from 0.70 to 0.90, revealing good internal consistency as all of them are 0.70 or above, which is the acceptable Cronbach's alpha value (O'Leary-Kelly and Vokurka, 1998; Wamba et al., 2020). These measures convey good item reliability and validity (Fornell and Larcker, 1981; Hair et al., 2007; O'Leary-Kelly and Vokurka, 1998; Wamba et al., 2020). As indicated by the fit indices, the model displayed a good fit with the data ($\chi^2 = 452.08$, $df = 236$, $\chi^2/df = 1.916$, TLI = 0.913, CFI = 0.925, RMSEA = 0.073). These measures,

along with their fit-indices above 0.90, indicate evidence of convergent validity (O'Leary-Kelly and Vokurka, 1998). Fornell and Larcker (1981) test, shown in Table 1, indicated that the square root of AVE of every construct is greater than the correlation between it and all other constructs, suggesting the presence of discriminant validity. The outer variance inflation factor (VIF) value ranges from 1.165 to 1.728 for interdependency, 1.939–2.872 for JPRS, 1.634–1.729 for innovation, 2.443–2.938 for SCSP, 1.799–2.253 for SCP, and 1.811–3.239 for SCEP, respectively. Values of inner VIF for all the proposed relationships ranged from 1.000 to 1.572. These measures indicate that multicollinearity is not a serious problem.

As a single method was used to collect data, we first attempted to reduce the common method bias (CMB) in data collection. Data confidentiality was ensured by conveying to respondents that no answer is right or wrong for getting an honest/unbiased response, and also promised to share aggregate results with respondents. Secondly, CMB tests were conducted as recommended by (Liang et al., 2007), which showed the average substantive variance explained by indicators is 0.706 while the average single method-based variance was 0.011. Further, Harman's (1976) tests also conveyed the worst fit compared to the research model. Thus, CMB is not a serious problem.

4.2. Structural model and moderation effects

First, direct relationships are assessed through the structural model. All direct links are found significant ($p < 0.001$ or $p < 0.01$), except JPRS→SCP ($p > 0.05$) and JPRS→SCSP ($p > 0.05$). Significant direct relationships are interdependency→JPRS ($\beta = 0.658$, $p < 0.001$), interdependency→innovation ($\beta = 0.306$, $p < 0.001$), JPRS→SCEP ($\beta = 0.291$, $p < 0.01$), innovation→SCEP ($\beta = 0.595$, $p < 0.001$), innovation→SCP ($\beta = 0.782$, $p < 0.001$), and innovation→SCSP ($\beta = 0.953$, $p < 0.001$).

To analyse the moderation effects, a multi-group analysis was conducted to understand the differences between moderator groups (low and high levels of moderator variables). Different resisting forces may change over time, so we studied their influence when they are at low and high levels. As the influence of all resistors on different relationships was tested individually, we explain the influence of each resistor on relationships leading to the TBL performance. As Steenkamp and Baumgartner (1998) recommended, this method is widely used in literature (Jajja et al., 2017; Wamba et al., 2020). With a five-point scale, cases that received 4 or higher were classified as *high*, and cases with scores less than 4 were classified as *low*. Invariance was tested across three dimensions: configural, measurement, and scalar invariance (Byrne, 2006). Models were tested with no constraints, equal factor loading constraints, equal factor loadings, and intercept constraints to test for configural, measurement, and scalar invariance. The structural invariance test was conducted by constraining all six (and eight) structural paths, one at a time, as equal across the groups, with all the above constraints present. Adequate model fit and change in CFI are considered to conclude evidence of noninvariance. A change of more than 0.01

Table 1
Correlations and square root of AVE in diagonal cells.

	DPS ^a	Innovation	Interdependency	JPRS	OB ^a	PA ^a	CI ^a	SCEP	SCP	SCSP
DPS	1.000									
Innovation	0.247	0.841								
Interdependency	0.352	0.431	0.902							
JPRS	0.433	0.603	0.564	0.795						
OB	0.472	0.096	0.183	0.241	1.000					
PA	0.421	0.051	0.197	0.227	0.655	1.000				
CI	0.657	0.294	0.305	0.378	0.371	0.460	1.000			
SCEP	0.376	0.599	0.480	0.711	0.207	0.318	0.403	0.873		
SCP	0.465	0.640	0.587	0.695	0.260	0.205	0.393	0.685	0.840	
SCSP	0.409	0.717	0.549	0.669	0.165	0.161	0.349	0.692	0.790	0.882

^a DPS: Difficulty in partner selection, OB: Opportunistic behavior, PA: Power asymmetry, CI: Conflicting interests.

(Cheung and Rensvold, 2002) or a change of more than 0.002 (Meade et al., 2008) or 0.001 (Byrne, 2010) are suggested for evidence of non-invariance, i.e., the presence of a moderation effect. To conclude non-invariance of barriers, we considered $\Delta CFI \geq 0.003$, which is neither a stringent nor a lenient criterion.

The results of the moderation effects are presented in Table 2. Although the fit-indices are not very high, considering a standard of 0.90, they are acceptable at moderate fit expectations (Forza and Filippini, 1998; Hair et al., 1995). For opportunistic behavior, the change in CFI (ΔCFI) for the configural model is 0.003 (for Innovation $\rightarrow$ SCSP and SCEP; JPRS $\rightarrow$ SCSP and SCEP) and 0.004 (for Innovation $\rightarrow$ SCP; JPRS $\rightarrow$ SCP). Thus, opportunistic behavior is observed to be moderating the relationships between innovation and TBL performance and between JPRS and TBL performance. In graphical form, moderation effects are presented in Fig. 2. A dot without a line in the graph indicates that a relationship is not significant at both high and low levels of the moderator. For example, the relationship between JPRS and SCSP is significant when power asymmetry is low and insignificant when it is high. Similarly, our results show that all relationship conditions, except power asymmetry, moderate the paths to achieve TBL-based performance from perceived interdependency, JPRS, and innovation. Hence, hypotheses, $H_{1a,1b}^-$, $H_{3(a-f)}^-$, $H_{4(a-f)}^-$ are supported while $H_{2(a-h)}$ related to

power asymmetry are not supported. These findings are further discussed in the discussion section.

Mediation effects were also analyzed. Mediation effects exist if the effects of the independent variables transmit through an intervening variable. All mediation or indirect paths, JPRS $\rightarrow$ Innovation $\rightarrow$ SCEP (0.343, $p < 0.01$), JPRS $\rightarrow$ Innovation $\rightarrow$ SCP (0.450, $p < 0.01$), JPRS $\rightarrow$ Innovation $\rightarrow$ SCSP (0.549, $p < 0.01$), and Perceived interdependence $\rightarrow$ JPRS $\rightarrow$ Innovation (0.379, $p < 0.01$) were found to be significant. As all direct paths, except JPRS $\rightarrow$ SCP and JPRS $\rightarrow$ SCSP, are also significant, innovation is partially mediating the relationship between JPRS and TBL dimensions. However, innovation fully mediates the relationship between JPRS $\rightarrow$ SCP and JPRS $\rightarrow$ SCSP.

5. Discussion and implications

To achieve sustainable development, collaboration and innovation are essential. However, there are numerous barriers in the form of relationship conditions that may act as impediments to collaboration and make the implementation of collaboration and innovation a difficult task. Taking insights from FFT and TCE, this research investigates moderating roles played by relationship conditions in achieving supply chain, environmental and social performance from JPRS and

Table 2
Moderation effect of relationship conditions.

Moderation effect of partner selection											
	χ^2	df	χ^2/df	TLI	CFI	RMSEA	Δdf	$\Delta \chi^2$	Sig. level	Low SRP	High SRP
1. Configural invariance	851.465	480	1.774	0.835	0.857	0.068					
2. Metric invariance	868.843	498	1.745	0.841	0.857	0.066	18	17.378	0.497		
3. Scalar invariance	917.068	522	1.757	0.839	0.847	0.067	42	65.603	0.011		
3a. PI [#] $\rightarrow$ Innovation	918.53	523	1.756	0.839	0.847	0.067	43	67.065	0.011	0.416*	0.163
3 b. PI $\rightarrow$ JPRS	929.562	523	1.777	0.834	0.843	0.068	43	78.097	0.001	0.762***	0.488***
Moderation effect of power asymmetry											
	χ^2	df	χ^2/df	TLI	CFI	RMSEA	Δdf	$\Delta \chi^2$	Sig. level	Low PA	High PA
1. Configural invariance	837.777	480	1.745	0.866	0.884	0.066					
2. Metric invariance	848.761	498	1.704	0.874	0.886	0.065	18	10.984	0.895		
3. Scalar invariance	875.526	522	1.677	0.879	0.885	0.063	42	37.749	0.658		
3a. Innovation $\rightarrow$ SCSP	876.985	523	1.677	0.879	0.885	0.063	43	39.207	0.637	0.789***	1.295**
3 b. Innovation $\rightarrow$ SCP	875.552	523	1.674	0.879	0.886	0.063	43	37.775	0.697	0.752***	0.579*
3c. Innovation $\rightarrow$ SCEP	879.24	523	1.681	0.878	0.884	0.063	43	41.462	0.538	0.495***	1.155**
3 d. JPRS $\rightarrow$ SCSP	878.538	523	1.68	0.878	0.885	0.063	43	40.761	0.569	0.191*	-0.396
3e. JPRS $\rightarrow$ SCP	875.538	523	1.674	0.879	0.886	0.063	43	37.761	0.697	0.208*	0.338
3f. JPRS $\rightarrow$ SCEP	883.867	523	1.69	0.876	0.883	0.064	43	46.09	0.346	0.453***	-0.457
3 g. PI $\rightarrow$ Innovation	875.791	523	1.675	0.879	0.885	0.063	43	38.013	0.687	0.285**	0.126
3 h. PI $\rightarrow$ Collaboration	875.54	523	1.674	0.879	0.886	0.063	43	37.763	0.697	0.671***	0.690**
Moderation effect of opportunistic behavior											
	χ^2	df	χ^2/df	TLI	CFI	RMSEA	Δdf	$\Delta \chi^2$	Sig. level	Low OB	High OB
1. Configural invariance	862.483	480	1.797	0.850	0.870	0.069					
2. Metric invariance	876.816	498	1.761	0.857	0.871	0.067	18	14.333	0.707		
3. Scalar invariance	912.84	522	1.749	0.859	0.867	0.067	42	50.357	0.176		
3a. Innovation $\rightarrow$ SCSP	914.161	523	1.748	0.859	0.867	0.067	43	51.678	0.171	0.963***	0.766***
3 b. Innovation $\rightarrow$ SCP	914.695	523	1.749	0.859	0.866	0.067	43	52.212	0.158	0.928***	0.526**
3c. Innovation $\rightarrow$ SCEP	912.924	523	1.746	0.860	0.867	0.066	43	50.441	0.203	0.549***	0.641***
3 d. JPRS $\rightarrow$ SCSP	913.418	523	1.746	0.859	0.867	0.066	43	50.935	0.190	0.001	0.205
3e. JPRS $\rightarrow$ SCP	915.256	523	1.75	0.859	0.866	0.067	43	52.773	0.146	0.024	0.422**
3f. JPRS $\rightarrow$ SCEP	913.041	523	1.746	0.860	0.867	0.066	43	50.558	0.200	0.347*	0.256a
Moderation effect of conflicting interest											
	χ^2	df	χ^2/df	TLI	CFI	RMSEA	Δdf	$\Delta \chi^2$	Sig. level	Low CI	High CI
1. Configural invariance	866.553	480	1.805	0.839	0.86	0.069					
2. Metric invariance	890.463	498	1.788	0.843	0.858	0.068	18	23.91	0.158		
3. Scalar invariance	934.504	522	1.79	0.842	0.851	0.068	42	67.951	0.007		
3a. Innovation $\rightarrow$ SCSP	937.11	523	1.792	0.842	0.85	0.068	43	70.557	0.005	1.101***	0.706***
3 b. Innovation $\rightarrow$ SCP	941.166	523	1.8	0.84	0.849	0.069	43	74.613	0.002	1.142***	0.374**
3c. Innovation $\rightarrow$ SCEP	935.374	523	1.788	0.843	0.851	0.068	43	68.821	0.007	0.762***	0.575**
3 d. JPRS $\rightarrow$ SCSP	937.181	523	1.792	0.842	0.85	0.068	43	70.628	0.005	-0.149	0.255*
3e. JPRS $\rightarrow$ SCP	943.354	523	1.804	0.84	0.848	0.069	43	76.801	0.001	-0.207	0.538***
3f. JPRS $\rightarrow$ SCEP	935.468	523	1.789	0.843	0.851	0.068	43	68.915	0.007	0.085	0.337**

#PI: Perceived interdependence, *** $p < 0.001$, ** $p < 0.01$, * $p < 0.05$, a $p < 0.10$. Shaded rows indicate significant moderating effects.

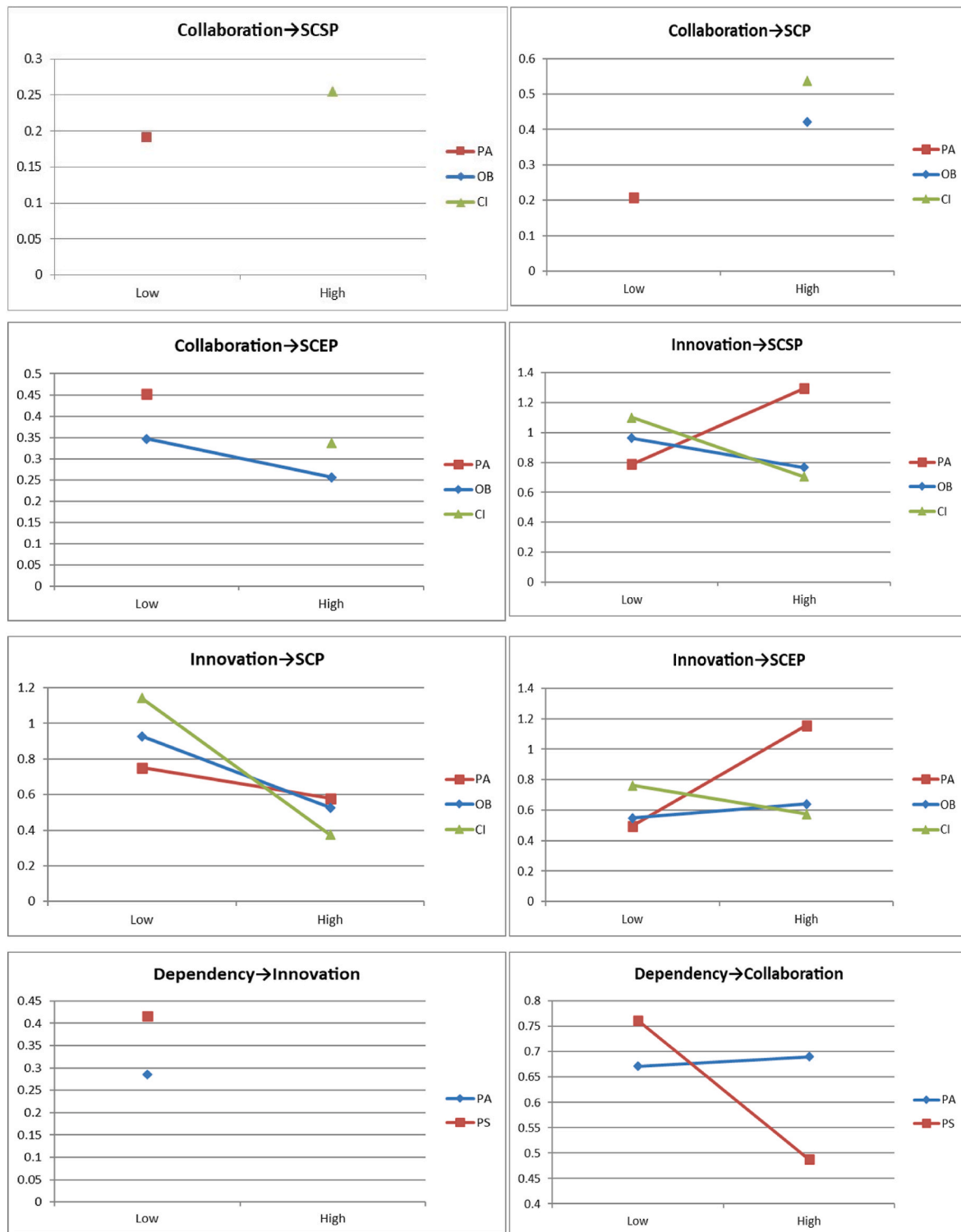


Fig. 2. Effects of moderation (a point on the graph indicates significant influence, and the absence of the point indicates insignificant influence).

innovation.

5.1. Direct effects on sustainability

The relationships, perceived interdependency to JPRS ($\beta = 0.658$, $p < 0.001$) and innovation ($\beta = 0.306$, $p < 0.001$) are supported. The more the perceived interdependency, the more quickly and intense will be JPRS and collaborative innovation. The perceived interdependency is more relevant in emerging economies like India, China, Brazil, and

Malaysia, where a lack of resources, use of child labor, poor working environments, and a lack of environmental awareness are more prominent (Govindan et al., 2014; Katiyar et al., 2018; Mani et al., 2016). So, managers of firms going to achieve SCS should identify the interdependency of partner firms.

The relationship, JPRS → SCEP ($\beta = 0.291$, $p < 0.01$) is supported, while the link JPRS → SCP ($\beta = 0.171$, $p > 0.05$) and JPRS → SCSP ($\beta = 0.009$, $p > 0.05$) were not supported. Our findings are aligned with studies in the US (Luzzini et al., 2015) and Brazil (da Silva et al., 2013)

that find collaboration to be beneficial for environmental performance.

Surprisingly, this study does not support the notion that collaboration or JPRS can enhance SCSP and SCP. In emerging economies, it is possible that JPRS itself is not sufficient for SCSP and SCP until firms utilize shared resources meaningfully by carrying out innovation and building dynamic capabilities (Kumar et al., 2018). As businesses and markets of emerging economies grow, firms strive to survive through competition rather than collaboration. As we also find that in the emerging economy, collaboration (Kumar, 2020; Kumar and Banerjee, 2014), innovation, and capabilities (McMahon and Thorsteinsdóttir, 2013; Santos et al., 2019) are not as advanced yet, a weaker collaboration may not lead to overall better SCP. Another reason could be that in the emerging economy, firms usually prioritize cost over quality, responsiveness (i.e., short lead time and customer satisfaction), and social responsibility-based outcomes. However, we measured SCP using both cost and responsiveness, so it resulted in unexpected results. Though firms spend on the health and education of the poor, donation, plantation, etc., as mandated by governments, they do not consider delivering impressive SCSP as an outcome of collaboration. Results of collaboration often fall short of expectations (Fadееva, 2005; Fawcett et al., 2015). Structural resistance and sociological make-up could be causes for firms not achieving relational capability (Fawcett et al., 2015), the primary capability of collaboration. However, firms benefit from collaboration even if the final goals are not achieved (Stuart, 2000).

The direct relationship between innovation and environmental ($\beta = 0.595$, $p < 0.001$), supply chain ($\beta = 0.782$, $p < 0.001$), and social ($\beta = 0.953$, $p < 0.001$) performance, which served as baseline expectations, were strongly supported. These findings convey to managers and decision-makers that to achieve SCS, innovation with supply chain partners is crucial. These findings complement existing literature (Dangelico and Pujari, 2010; Gürelek and Tuna, 2018) in that innovation is required to develop and deliver green products. Nonetheless, our empirical findings convey that sustainability-oriented innovation is positively linked to social, supply chain, and environmental performance. As emerging economies have many infrastructure, health, education, and poverty-related problems, the results suggest to managers that collaborative innovation is crucial to effectuate SCS, including social innovation, to address social-ills and requirements (Herrera, 2015; Mirvis et al., 2016).

5.2. Relationship conditions as moderators

A look at behaviors of perceived interdependency-JPRS-innovation-TBL performance linkages across different levels of relationship conditions suggests that the behavior of these linkages is complex.

5.2.1. Partner selection

Our results show that hypotheses H_{1a} and H_{1b} are supported. This implies that partner selection moderates the interdependency-JPRS and interdependency-innovation links. Greater difficulties in selecting partners significantly reduce the chance of initiating JPRS (see Fig. 2). These observations convey to managers that firms may still collaborate in the absence of suitable supply chain partners, but their chances and intensity are considerably reduced. When finding compatible partners is difficult, liaising with the right partners with complementary capabilities and aligned goals is difficult. In such cases, managers may practice less intense collaboration.

Similarly, greater difficulties in selecting the right partners make the interdependency-innovation link insignificant (see Fig. 2). It suggests to managers that having a suitable or compatible partner is required for innovation. Managers in emerging economies with limited resources may see high risks in investing it for innovation if identifying a compatible partner is difficult. Partners may be unsuitable if their goals are not aligned, if they do not possess enough capabilities for innovation and sustainability, and if there is a high risk for opportunistic behavior.

These factors work against effective innovation. Hence, the difficulty associated with finding the right partners constrains the involvement in collaboration and innovation, which will further jeopardize TBL's performance.

5.2.2. Power asymmetry

Our results show that power asymmetry does not moderate any of the relationships. Therefore, hypotheses H_{2a} to H_{2h} are not supported. Although this result was unexpected, the findings suggest to managers of emerging economies that if interdependency is understood, power asymmetry is immaterial in going for collaboration and innovation and subsequently achieving SCS. It suggests to managers that firms with power asymmetry can collaborate and innovate to achieve SCS. It is possibly because firms in emerging economies are more interdependent for innovation (Kothiyal, 2018; Yourstory, 2020) and SCS, and hence they do not exercise their power to exploit partner firms. Our findings are different from the literature (Cho et al., 2019; Touboulic et al., 2014) that suggests power asymmetry is an inhibitor of collaboration and also the literature (Michalski et al., 2017) that suggests powerful partners or institutions can enforce protocols and standards to achieve compliance (Chen and Kitsis, 2017).

5.2.3. Opportunistic behavior

The results reveal that a higher degree of opportunistic behavior reduces the effectiveness of the relationships between JPRS and SCEP, between innovation and SCP, and between innovation and SCSP, thereby supporting hypotheses H_{3a} , H_{3e} , and H_{3f} . These findings support previous literature (Cheng and Sheu, 2012) in that the higher the opportunistic behavior within collaborative alliances, the less favorable the alliance outcomes. However, hypotheses H_{3b} , H_{3c} , and H_{3d} were not supported.

The effectiveness of the JPRS-SCEP link depends more on shared information, and the leakage of such information impacts the link negatively. The finding conveys to managers that firms in emerging economies are vulnerable to opportunistic behaviors and leakage that sabotage SCEP. However, the JPRS-SCP link is significant only at high levels of opportunistic behavior. It may be because firms in emerging economies are so interdependent on information and resources that even in the presence of opportunistic behaviors, collaboration and innovation bring better SCP. It suggests that managers should control opportunistic behaviors and go ahead to practice collaboration and innovation (Yourstory, 2020) to enhance SCS. Surprisingly, the JPRS-SCSP link is observed to be insignificant. However, JPRS interestingly is shown to be impacting SCSP through innovation.

An opportunistic partner may subvert alliance goals to obtain short-term benefits, which reduces the desired outcomes. For SCP, if technology, delivery strategy, inventory control methods, information for innovation, such as forecast, lead time, delivery time, inventory management system, routing mechanism, customer complaints, and other ways to deal with supply chain problems, are leaked, competitors, as well as partner, may use it to their advantage. A better ability of competitors to satisfy customers will result in reduced market share. If the manufacturer knows their supplier's future plan, the manufacturer may reduce orders, or they may differ order time in anticipation of getting a better deal in the coming future, which will impact overall SCP negatively (Tan et al., 2016). If crucial product innovation information has been leaked to competitors, competitors may use it to grab more market share, reducing SCP and SCSP. Hence, such leakages will influence the innovation-SCP link negatively. As such, for socially relevant innovation, sharing the required information about employees and society is crucial, and partners may use such information in negotiation to get more individual benefits which will reduce SCP. It will also reduce SCSP from innovation as a partner in opportunistic behavior situations may refrain from sharing socially relevant information. Suppliers may hide child labor and poor work conditions in their production plants; however, such shared information for innovation may again be leaked by

partners. Such leakage can be exploited by competitors, eventually leading to defamation and poor social performance. Competitors will likely benefit from the leaked information without initiating collaborative relationships with their supply chain members.

Opportunistic behavior is found to be moderating the innovation-SCEP link positively. Compared to SCSP and SCP, SCEP is more technology sensitive as it requires eco-friendly products and processes. SCEP in emerging economies is largely a less focused performance dimension, so it is possible that even if valuable information is leaked or misused, collaborative alliances are still able to improve their SCEP as they have access to valuable, required shared resources. In the long term, however, competitors may be able to perform relatively better based on such leaked or misused information. Hence, in an emerging economy context, where the extent of eco-friendly operations and technology is poor, managers can form collaborative alliances to improve the innovation-SCEP link, even in the presence of opportunistic behavior. However, managers should formulate a foolproof mechanism to control opportunistic behavior that can help achieve excellent SCS.

5.2.4. Conflicting interest

Finally, the moderating effects of conflicting interests on the innovation-TBL link (H_{4d} , H_{4e} , H_{4f}) are supported. However, there is no evidence for the negative moderating effects of conflicting interest on the JPRS-TBL link; it rather appears to be moderating favorably. Therefore, H_{4a} , H_{4b} , H_{4c} are not supported.

An independent sample test on JPRS (and innovation) factor score and conflicting interests (high and low) was conducted to understand it further. We found mean score of JPRS (mean difference = 0.60, $p < 0.05$) and innovation (mean difference = 0.411, $p < 0.05$) are significantly higher at a high level of conflicting interest than the mean score of JPRS and innovation at a low level of conflicting interest. This test and Fig. 2 convey to managers that conflicting interest poses more problems to the innovation-TBL link than the JPRS-TBL link. Innovation as a joint activity requires intense interaction, precious information and resource sharing and their transformation, which naturally will invite more conflicts (Vollmer, 2015). This finding conveys to managers that conflicting interests negatively influence the innovation-TBL link if not resolved on time. As innovation is crucial for TBL performance in collaborative settings, a decline in the effectiveness of the innovation-TBL link will be counterproductive in improving TBL performance.

Interestingly, at higher levels of conflicting interests, all JPRS-TBL links are significant, but at lower levels of conflicting interests, the JPRS-TBL links are not significant. As managers in emerging economies face resource and capability scarcity, higher collaboration improves SCS performance even with higher conflicting interests. Operationally, when alliances have deeper levels of collaboration, they share valuable complementary resources and information, discuss issues together, learn from each other, and interact intensely, which may create more conflicts. However, at deeper levels of collaboration, desired outcomes of collaboration surpass the negative impacts of higher conflicting interests. This is also in line with the literature (Stuart, 2000) in that resultant outcomes are beneficial despite numerous barriers and difficulties in implementing collaboration. Reaching a consensus may not always mean the disappearance of conflicts, but deeper or conflict-prone problems are sacrificed for the benefit of consensus and greater goals of collaboration (Fadeeva, 2005). At the same time, evidence shows conflicts encouraging collaboration initiatives and associated goals (Fadeeva, 2005). Hence, a positive approach toward conflicts may not always degenerate sustainable collaborative outcomes. Our findings convey to managers in emerging economies that they should not fear conflicting interests excessively, deploy conflict resolution mechanisms, and practice collaboration and innovation at a deeper level to improve SCS.

5.3. Practical implications

This research has unique findings and important practical and policy implications to offer. We find that perceived interdependence motivates firms to collaborate and to innovate to enhance social, environmental, and SCP. Thus, managers and policy-makers should invest their time and efforts to identify supply chain partners who are more interdependent. Dependent partners may keep working in exploitative relationships (Johnsen and Lacoste, 2016). As dependent partners may keep collaborating and innovating, policy-makers can recommend long-term honest relationships for a larger benefit.

JPRS may not directly result in enhanced performance across all three SCS dimensions. However, collaboration influences TBL's performance through innovation. JPRS influences SCEP directly as well as through innovation. Innovation further mediates the relationship of JPRS to SCSP and JPRS to SCP. It conveys to the managers that not only collaboration but collaborative innovation is also crucial to enhance SCS. While collaborating, managers and policy-makers can have the policy to carry out joint innovation to improve environmental, social, and SCP. If a policy of innovation is not in place, the potential of collaborative relationships in terms of TBL performance may not be realized.

Power asymmetry does not moderate the relationships, while difficulty in partner selection creates hurdles for collaboration and does not allow collaborative innovation. It first conveys to managers and policy-makers that power asymmetry may not always be a negative force in forming sustainable collaborative partnerships. However, they require an effective mechanism to reduce difficulties in finding suitable partners with aligned goals for collaboration and innovation to improve SCS. While in the collaborative relationship, managers and policy-makers should be vigilant about if the powerful partner does not exercise their power to threaten or sanction other partners, it may not bring negative consequences for TBL performance. They can also be aware of the positive power that can be used for cooperation, attaining common goals (Johnsen and Lacoste, 2016), and compliance.

Interestingly, higher opportunistic behavior displays mixed results in deteriorating SCS. It brings a deteriorating impact on the relationship of innovation to SCSP, innovation to SCP, and JPRS to SCEP links. However, innovation to SCEP and JPRS to SCP links are strengthened. Managers and policy-makers should know that opportunistic behaviors in all forms may not bring negative TBL performance. Collaborative innovation especially seems very effective, where some opportunistic behaviors in certain forms may not damage SCEP, SCSP, and SCP. Little opportunistic behavior may not impact collaboration to TBL performance that is different from the literature (Cheng and Sheu, 2012). However, it is more harmful to innovation and its corresponding TBL performance. Managers and policy-makers, thus, should safeguard innovative activities more from opportunistic behaviors to ensure better SCP and SCSP. It also conveys to managers that if collaboration is practiced along with innovation, the overall negative impact of opportunistic behavior can be reduced, and SCS can be improved.

As conflicting interests deteriorate the innovation-TBL link, managers and policy-makers of firms interested in improving SCS should reduce conflicting interests in innovation. Managers and policy-makers should be careful about conflicting interests that are potentially damaging to innovation driving TBL performance. Further, we find that higher conflicting interests may not weaken the JPRS-TBL link. Hence managers should carry on collaboration even if conflicting interests exist. A certain degree of conflict may maintain partners driving force and may be acceptable for creativity, learning, value creation efforts (Johnsen and Lacoste, 2016; Matsuo, 2006), and TBL performance. Managers and policy-makers should also have a positive view of conflicts implying it always does not damage the relationship; instead, it can tighten the relationship (Johnsen and Lacoste, 2016), leading to better TBL performance.

In summary, this study suggests managers and policy-makers that

innovation and collaborative relationships are crucial for achieving TBL performance in emerging economies like India. Collaborative innovation is so important that even in the presence of some negative elements of the relationship atmosphere, it will help realize good TBL performance. At the same time, managers and policy-makers should control such negative elements to realize the full potential of collaborative innovation in the supply chain. This research contributes to the body of knowledge by exploring how firms in emerging economies, specifically India, pursue collaboration and innovation to achieve SCS, including social, environmental, and supply chain dimensions.

5.4. Theoretical implications

This study illuminates FFT and TCE in the context of TBL performance. Only collaboration does not appear effective in improving SCS. The presence of an innovation construct makes collaboration more effective for TBL performance. In achieving improved TBL performance, innovation can utilize, reconfigure, and control joint resources more effectively and efficiently, which will increase asset specificity in terms of TBL performance and reduce transaction costs. Hence, in this context, TCE can be understood as encouraging collaborative innovation for better TBL performance.

According to FFT, difficulties in finding suitable partners prevent firms from going from the present state to the next state, i.e., toward more collaborative relationships. Difficulty in finding suitable partners is an important barrier to go for collaborative innovation.

To achieve SCS, power asymmetry neither resists changes in firms' state, as FFT suggests, nor does it seem to be a barrier while in a collaborative relationship, as TCE suggests. Power asymmetry, thus, does not appear to be a negative relationship condition for SCS. Power asymmetry may not result in negative consequences if power is not used to force the partners. Our findings are in line with a stream of previous literature (Johnsen and Lacoste, 2016; Kähkönen, 2014; Nyaga et al., 2013; Zhuang et al., 2010) that views non-mediated power as unharmed. With non-mediated power, customers (or buyers) may be powerful, but they may not use this power against their suppliers (Johnsen and Lacoste, 2016; Kähkönen, 2014). Our findings differ from previous literature (Hall and Matos, 2010; Johnsen and Lacoste, 2016; Michalski et al., 2017) that perceives power asymmetry as dysfunctional for the relationship.

Considering TCE, opportunistic behavior displays mixed results and does not always impose high transaction costs in improving SCS. It creates a more deteriorating impact on innovation-based TBL performance (except SCEP), while collaboration-based TBL performance displays a negative impact only for SCEP. As SCEP requires more technical and technological support, opportunistic behavior is unable (able) to damage SCEP driven by innovation (simply collaboration). Our results are partially in line with previous literature (Cheng and Sheu, 2012; Mohr and Spekman, 1994) that says opportunistic behavior can make relationships dysfunctional.

From a TCE lens, conflicting interest also displays complex behavior. For the innovation-TBL performance link, conflicting interest reduces the final outcomes of TBL performance. For innovation, conflicting interests impose additional transactional costs that make innovation inefficient, which in turn subverts TBL performance. This finding is in line with previous literature (Cheng and Sheu, 2012; Johnsen and Lacoste, 2016; Zhuang et al., 2010) that suggests conflict would destabilize the relationship. However, it is also observed that at low levels of conflicting interest, the JPRS-TBL performance links are not significant, while they are significant at high levels of conflicting interest. This observation is not aligned with TCE. This finding possibly explains the interplay of JPRS and conflicting interests, with the aggregate outcome of JPRS and conflicting interests at different JPRS levels being different. Our findings support previous literature (Johnsen and Lacoste, 2016; Matsuo, 2006) that conflict does not always play a negative role but has a dual nature that can impact positively as well as negatively.

This study is perhaps the first to investigate the role of different relationship conditions between collaboration-innovation and TBL performance. Difficulties in partner selection, power asymmetry, opportunistic behavior, and conflicting interests are considered negative relationship conditions. Moderating effects of relationship conditions on the linkages of interdependence-collaboration-innovation-TBL performance are analyzed and explained in the context of FFT and TCE.

6. Conclusions

The present research proposed collaborative innovation and analyzed relationship-based boundaries under which SCS can be realized. Collaboration-innovation-TBL performance linkages depict complex behaviors based on the relationship conditions as contingency variables. These variables act as boundaries to explain collaboration-innovation-TBL linkages in the supply chain. Our major findings are as follows.

1. The difficulty associated with selecting partners for collaboration reduces the depth of collaboration, while firms may not prefer carrying out innovation in such cases.
2. Power asymmetry does not impose significant transactional costs and therefore does not negatively influence the relationship between collaboration and innovation to TBL performance.
3. Opportunistic behavior displays complex results in realizing TBL performance. It does not consistently act as a source of transactional costs for both innovation and collaboration-based TBL performance.
4. The problem of conflicting interests results in a negative impact on the innovation-TBL link. However, more conflict results in an improved collaboration-TBL link.

Recognizing the importance of perceived partner interdependency, collaboration, innovation, and sustainability in emerging economies, this study sought to advance research on sustainable supply chain management by building a model based on FFT and TCE. The importance of innovation in collaborative relationships was highlighted, and the behavior of the interdependency-JPRS-innovation-TBL linkage under different relationship conditions was analyzed. We investigated how different negative relationship conditions moderate interdependency-JPRS-innovation-TBL relationships.

This research has unique findings and important practical and theoretical implications that have been discussed in the previous section. We find that perceived interdependency motivates firms to collaborate and innovate to enhance social, environmental, and SCP. JPRS may not directly result in enhanced performance across all three SCS dimensions. However, the JPRS/collaboration influences sustainability performance through innovation. JPRS influences SCEP directly as well as through innovation. Innovation is required for the effectiveness of both SCSP and SCP. It conveys to the managers of different businesses that collaboration and collaborative innovation are also important to enhance SCS.

Power asymmetry does not moderate the relationships, while difficulty in partner selection creates hurdles for collaboration and does not allow collaborative innovation. It conveys to managers that, for sustainable collaboration and innovation, they can ignore the power asymmetry of their partners. Still, they may require an effective mechanism to reduce difficulties in finding suitable partners with aligned goals for collaboration and innovation to improve SCS. In general, managers should reduce opportunistic behavior to strengthen the relationship of collaboration and innovation to TBL performance. Interestingly, higher opportunistic behavior displays mixed results in deteriorating SCS. As conflicting interests decline the innovation-TBL link, managers of firms interested in improving SCS should reduce conflicting interests in innovation. Further, we find that higher conflicting interests may not weaken the JPRS-TBL link. Hence managers may carry on collaboration even if conflicting interests exist. This

research contributes to the body of knowledge by exploring the way firms in emerging economies, specifically India, pursue collaboration and innovation to achieve SCS, including social, environmental, and supply chain dimensions.

6.1. Contributions and future scopes

This study makes important theoretical contributions in the domain of SCS. Our findings are related to the discussion of collaboration and innovation to improve the prospects of SCS (Arora et al., 2020; Chen and Kitsis, 2017; Cheng and Sheu, 2012; Hall and Wagner, 2012; Wiengarten and Longoni, 2015). To the best of our knowledge, this is the first study that uses FFT and TCE theories to understand improving interdependency-collaboration-innovation-TBL behaviors by considering moderating roles of different negative relationship conditions. Understanding nuances of negative relationship conditions and collaborative innovation for SCS is the primary theoretical contribution of this study to the related literature (Melandar and Pazirandeh, 2019; Michalski et al., 2017; Oyedijo et al., 2021). We explored a few of those negative relationship conditions that are mainly understood as inhibitors to collaboration and, subsequently, to sustainability (Cheng, 2011; Cheng and Sheu, 2012; Nyaga et al., 2013; Sancha et al., 2016). Such nuances are explored in an emerging economy, adding value to the sustainability debate in emerging economies (Kumar et al., 2018; McMahon and Thorsteinsdóttir, 2013; Silva et al., 2021). The finding that innovation fully mediates the relationship between collaboration and dimensions of SCS (except SCEP) further adds important value to the sustainability debate in emerging economies. Such understanding is critical and timely in emerging economies where firms have just begun putting efforts into addressing sustainability concerns (Jayaram and Avittathur, 2015).

Although, we believe that our theoretical model is generalizable, it needs further validation. Case studies should be carried out to unearth in-depth phenomena about the roles of barriers to collaboration in nurturing the collaboration-innovation-TBL link. It will also be beneficial to validate if the same behaviors related to the perceived interdependency-collaboration-innovation-TBL linkages are observed across different industries of developing and developed economies. A longitudinal study will add value to our proposed model as we draw from FFT. The state of negative relationship conditions may change over time, and the influence of their interaction on the collaborative relationship can be studied in the future. Our findings also motivate future research to explore different forms of negative relationship conditions, such as relating to relationship, production, process, technology, product, etc. Future studies should also explore the development of such negative relational elements-based models for TBL performance using other theories, such as institutional, relational, social network, contingency, etc. Contingent specificities, such as gas price, technological environment, different country regulations, etc., and industries specificities (as the food supply chain is different from the automobile supply chain) are important for sustainability that should be explored in future studies to understand SCS performance better.

Besides collaboration and innovation, dynamic capabilities are also required to achieve sustainability. It needs to be seen if dynamic capabilities and relational capital bring some new aspects of understanding, which may be useful for both developed and emerging economies. We used a limited number of measurement variables; future studies should use more variables to study specific negative relationship conditions as inhibitors to collaboration-sustainability-performance linkages. Especially, including flexibility in SCP, resource/energy reduction in SCEP, and measuring relationship conditions using multiple items will add more value. As we call for more research on the perceived interdependency-collaboration-innovation-TBL linkage, we believe the presented study acts as a building block for exploring the negative relationship atmosphere for TBL performance, which is extendable to other developing and developed economies.

Data availability

Data will be made available on request.

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